Ministerul Educației și Cercetării al Republicii Moldova
Universitatea Tehnică a Moldovei
Facultatea Calculatoare, Informatică și Microelectronică

Raport
la
Lucrarea de laborator Nr. 4

Disciplina: Tehnici de programare

Tema: Python

A efectuat : st. gr.SI – 264 Railean Adrian

A verificat : asistent universitar Reițman Patricia

Chișinău – 2026

SARCINA LUCRĂRII

Aprofundarea sintaxei și a conceptelor din limbajul Python prin rezolvarea exercițiilor pe platformele Codedex.io și Learnpython.org:

- Operații de bază: afișare text, variabile, operatori aritmetici și formule matematice.
- Structuri de date și control: liste, dicționare, seturi, condiții și bucle.
- Concepte avansate: funcții, clase/obiecte, expresii regulate, tratarea excepțiilor, generatori și serializare JSON.

1) Codedex.io/python:

Hello World

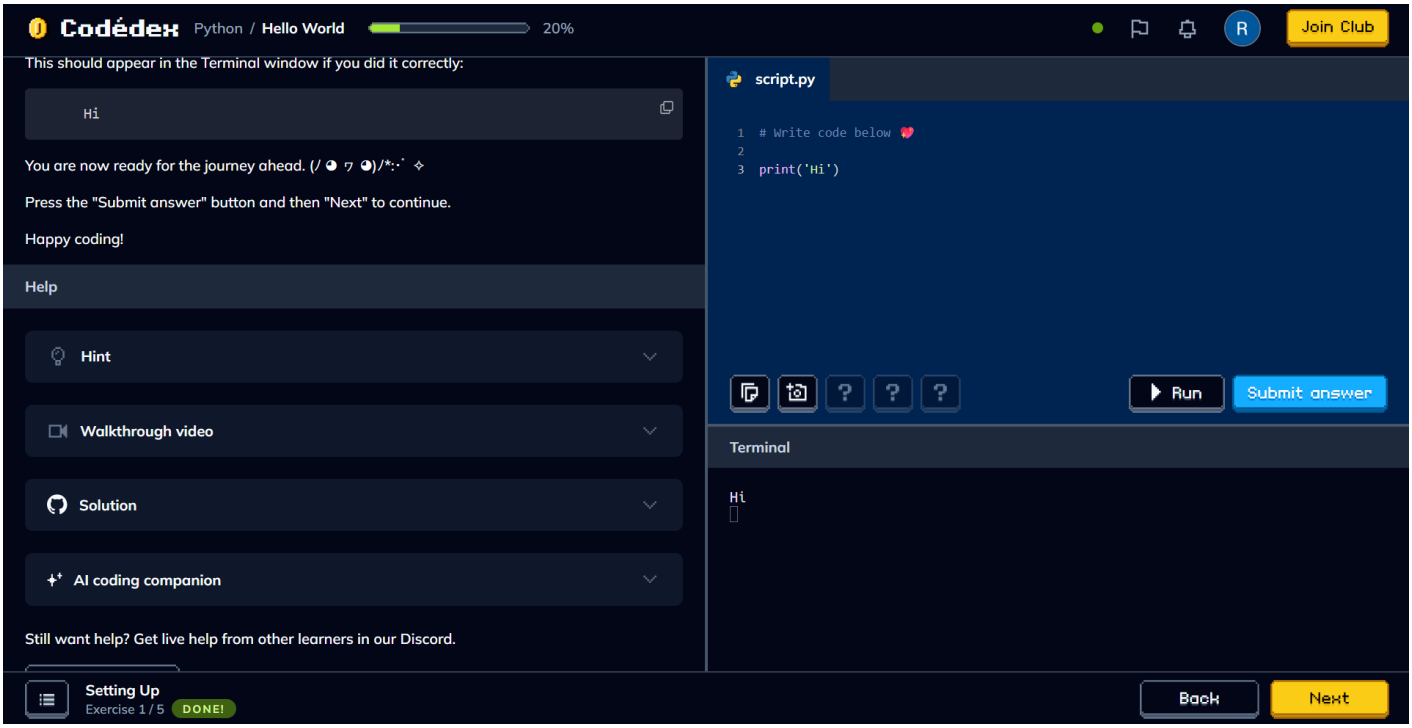


Figura 1.1

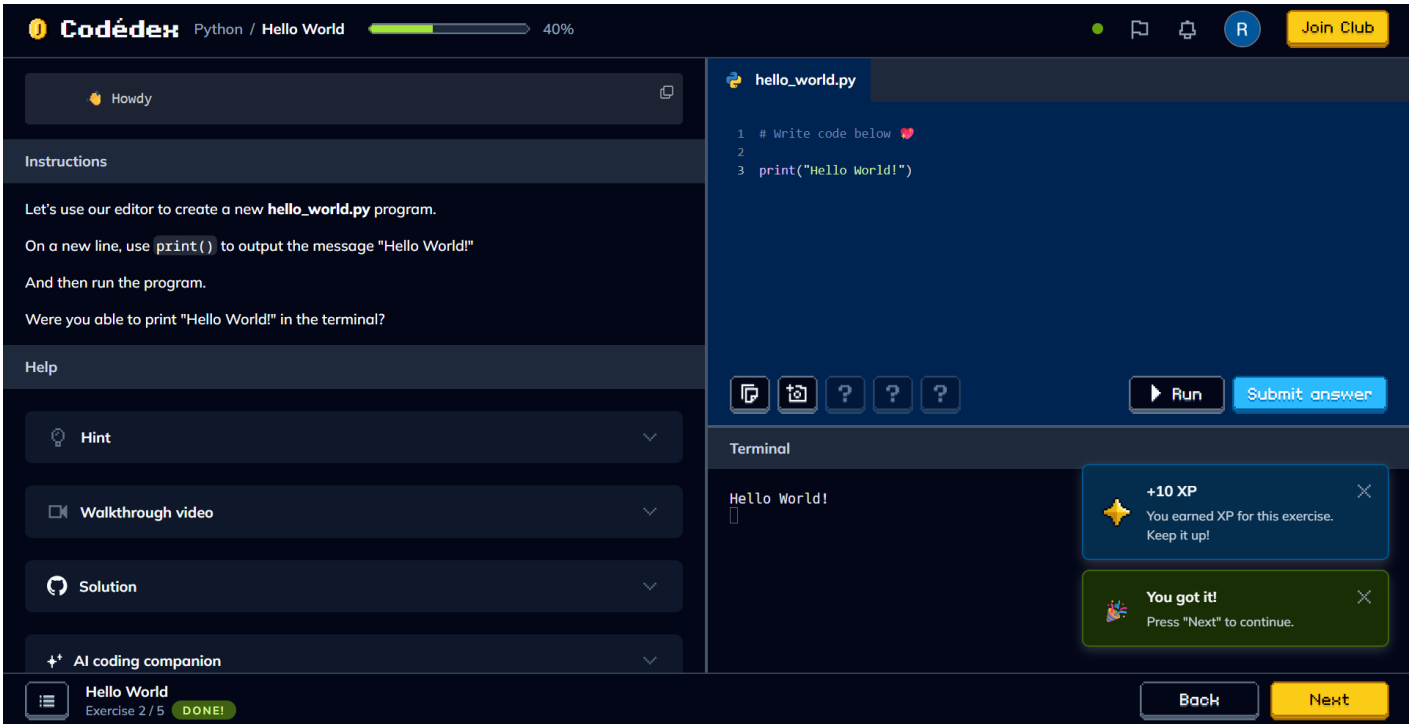


Figura 1.2

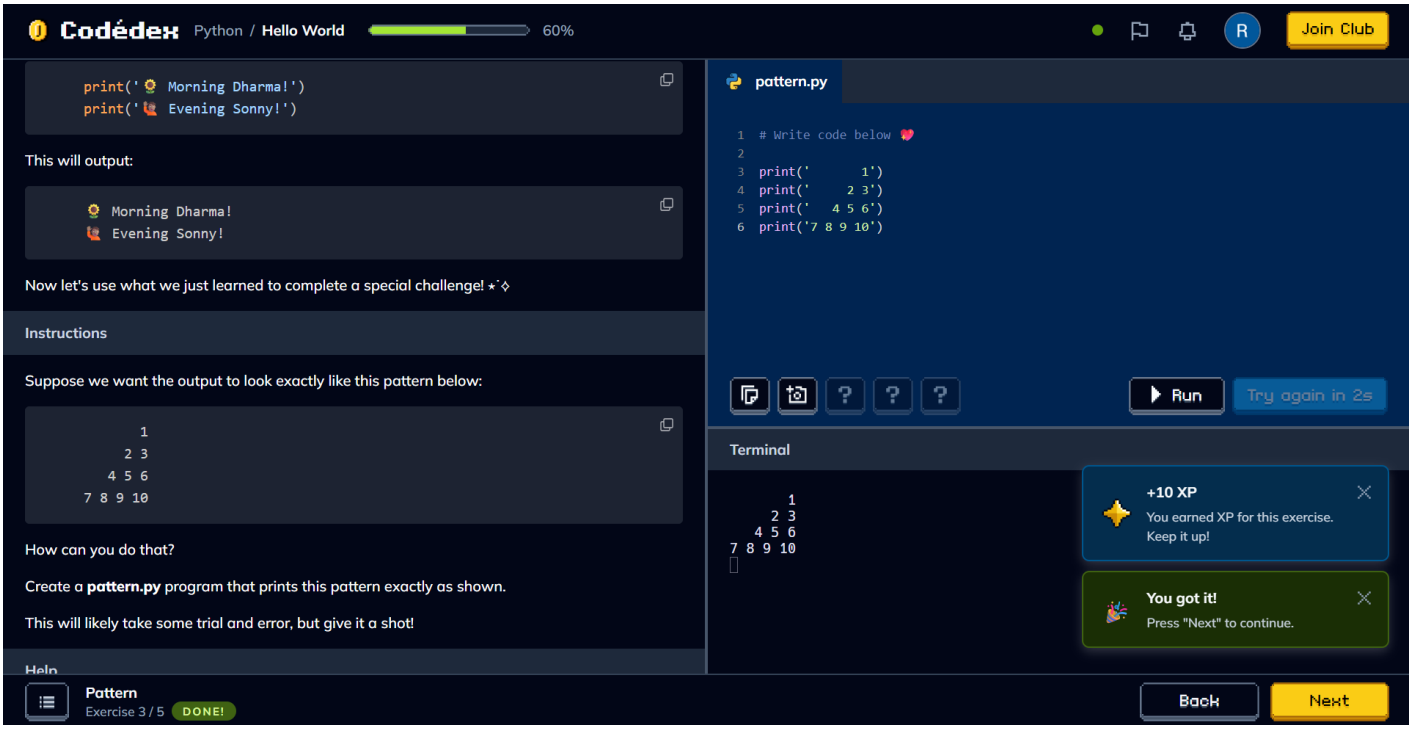


Figura 1.3

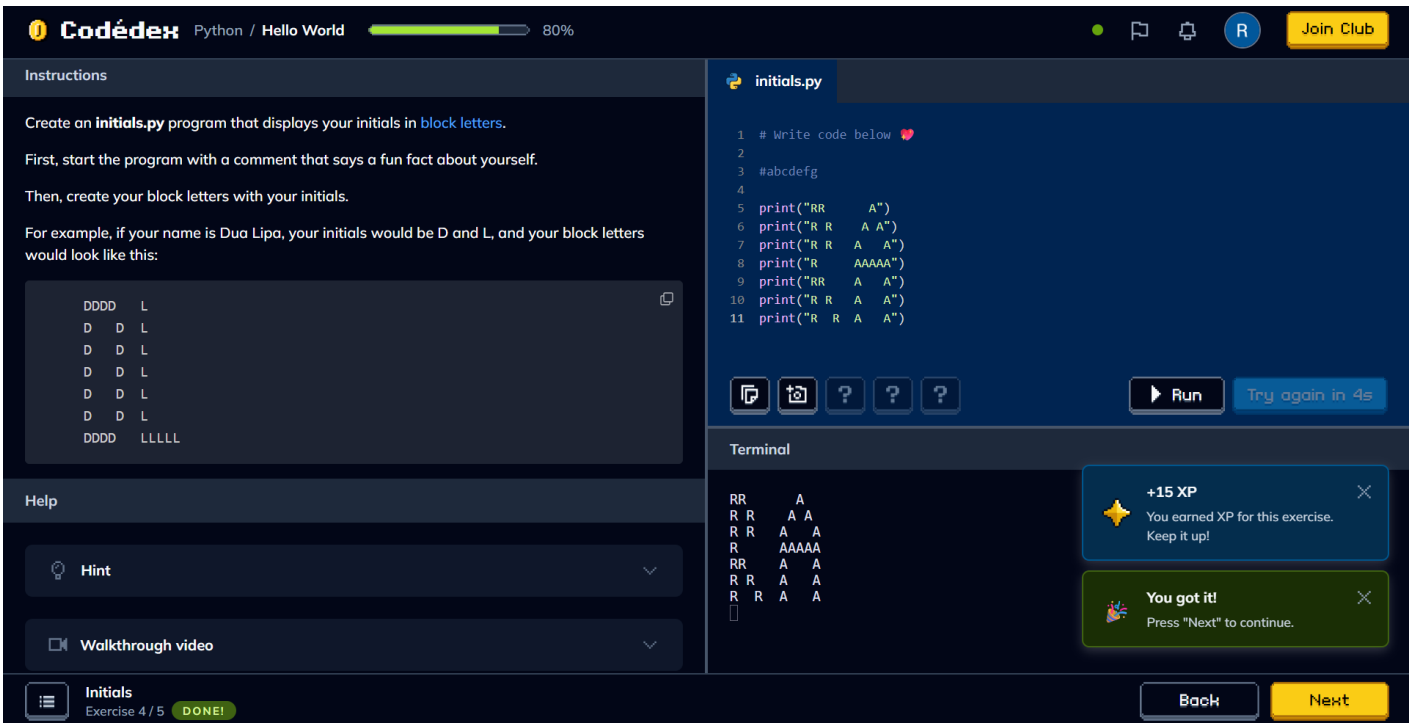


Figura 1.4

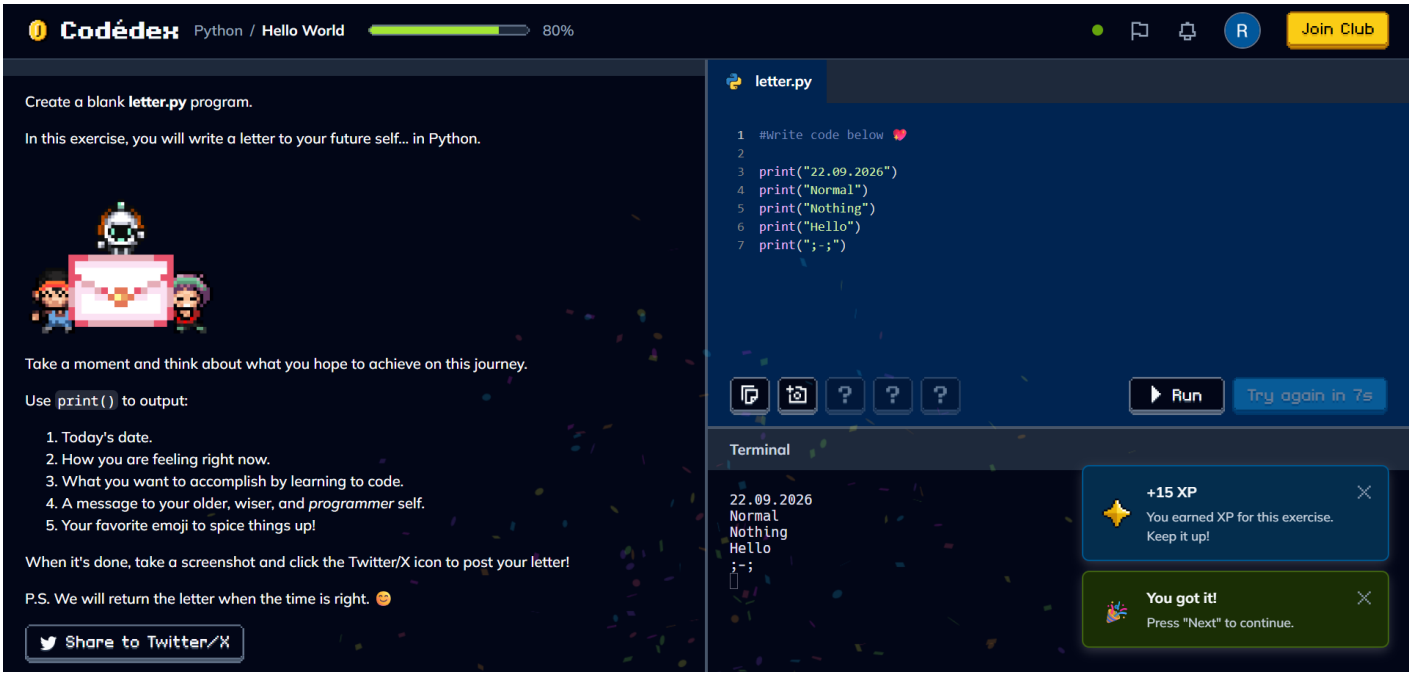


Figura 1.5

Variables

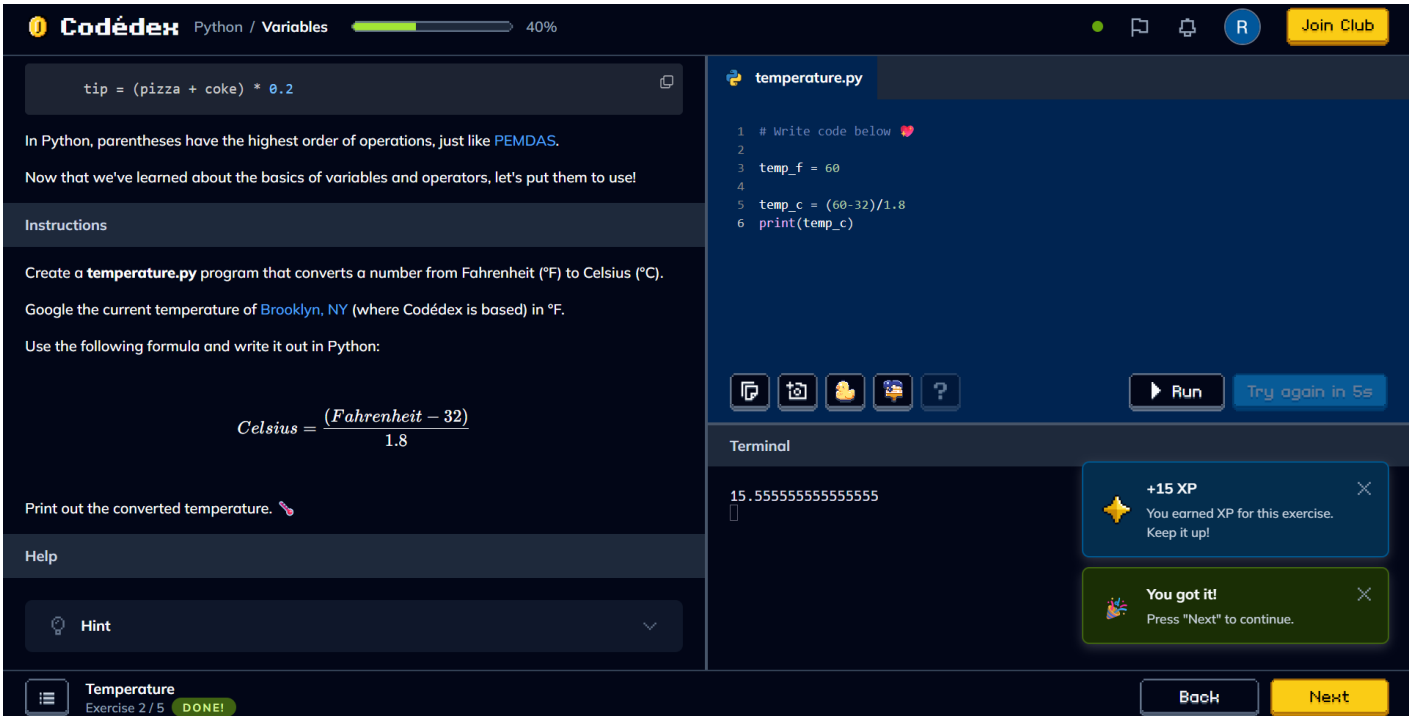


Figura 1.6

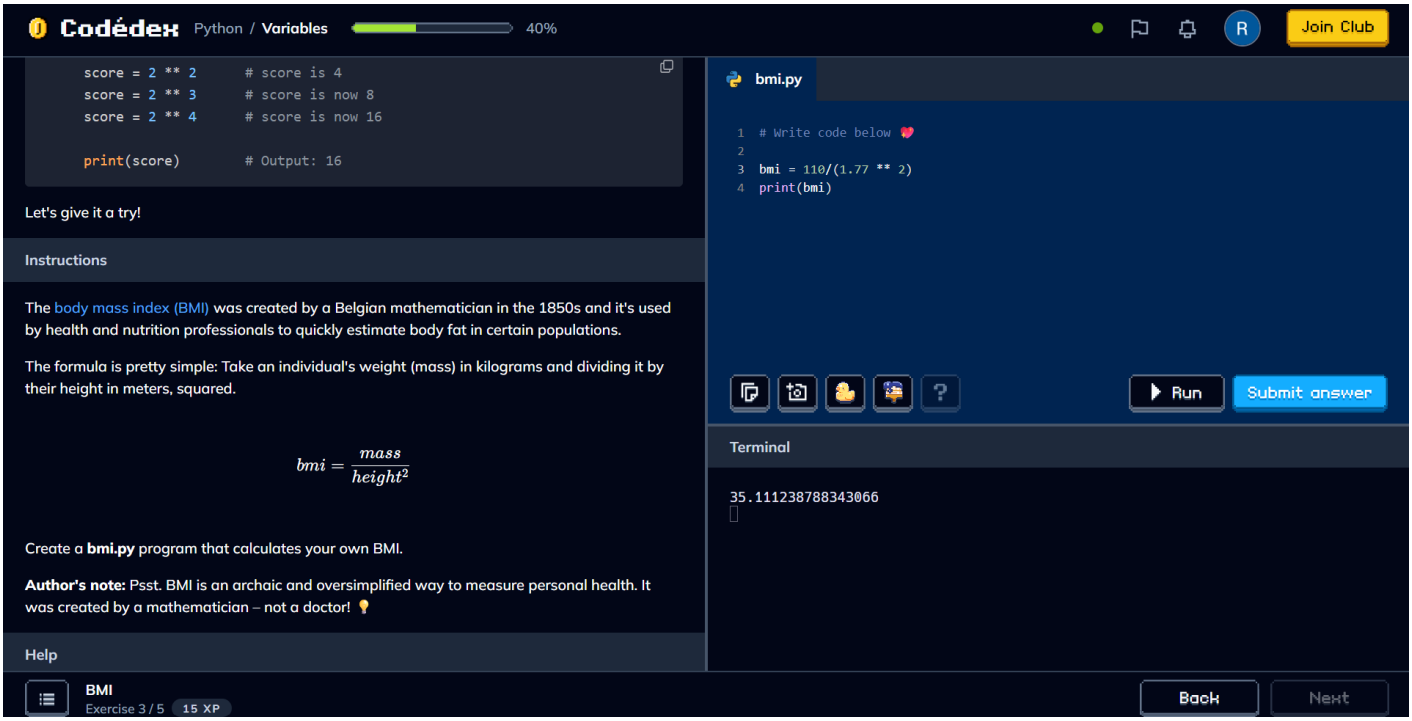


Figura 1.7

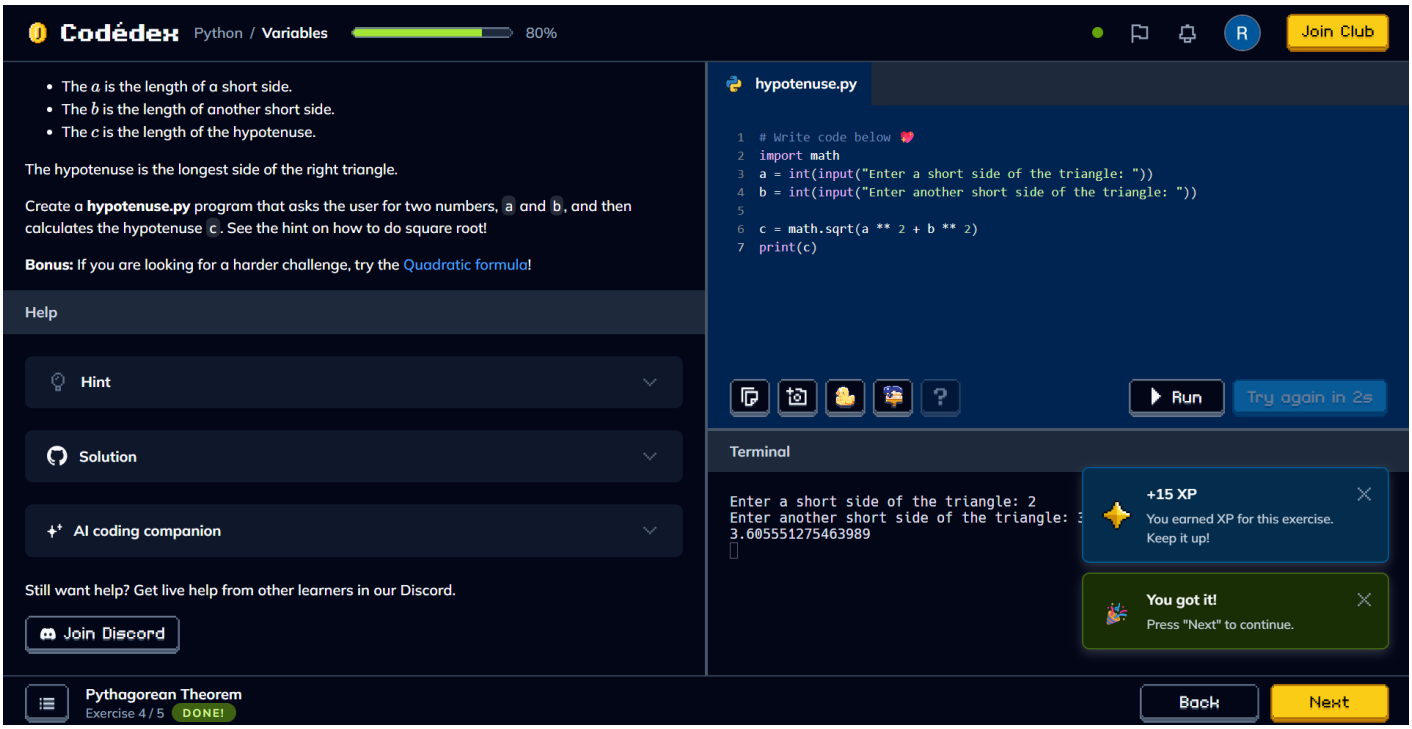


Figura 1.8



Figura 1.9

2) Learnpython.org:



Figura 2.1 – Hello World

Exercise

The target of this exercise is to create a string, an integer, and a floating point number. The string should be named `mystring` and should contain the word "hello". The floating point number should be named `myfloat` and should contain the number 10.0, and the integer should be named `myint` and should contain the number 20.

script.py

```
1  # change this code
2  mystring = "hello"
3  myfloat = float(10)
4  myint = 20
5
6  # testing code
7  if mystring == "hello":
8      print("String: %s" % mystring)
9  if isinstance(myfloat, float) and myfloat == 10.0:
10     print("Float: %f" % myfloat)
11 if isinstance(myint, int) and myint == 20:
12     print("Integer: %d" % myint)
```

IPython Shell

```
<script.py> output:
    String: hello
    Float: 10.000000
    Integer: 20

In [1]:
```

Great job!

Solution | Submit

Figura 2.2 – Variables and Types

script.py | **solution.py**

```
1  numbers = []
2  strings = []
3  names = ["John", "Eric", "Jessica"]
4
5  # write your code here
6  second_name = names[1]
7
8  numbers.append(1)
9  numbers.append(2)
10 numbers.append(3)
11
12 strings.append("hello")
13 strings.append("world")
14
15 # this code should write out the filled arrays
   and the second name in the names list (Eric).
```

IPython Shell

```
Traceback (most recent call last):
  File "script.py", line 12, in <module>
    strings.append("hello", "world")
TypeError: append() takes exactly one argument (2 given)

<script.py> output:
    [1, 2, 3]
    ['hello', 'world']
    The second name on the names list is None

<script.py> output:
    [1, 2, 3]
    ['hello', 'world']
    The second name on the names list is Eric
```

Great Job!

Submit

Figura 2.3 – Lists

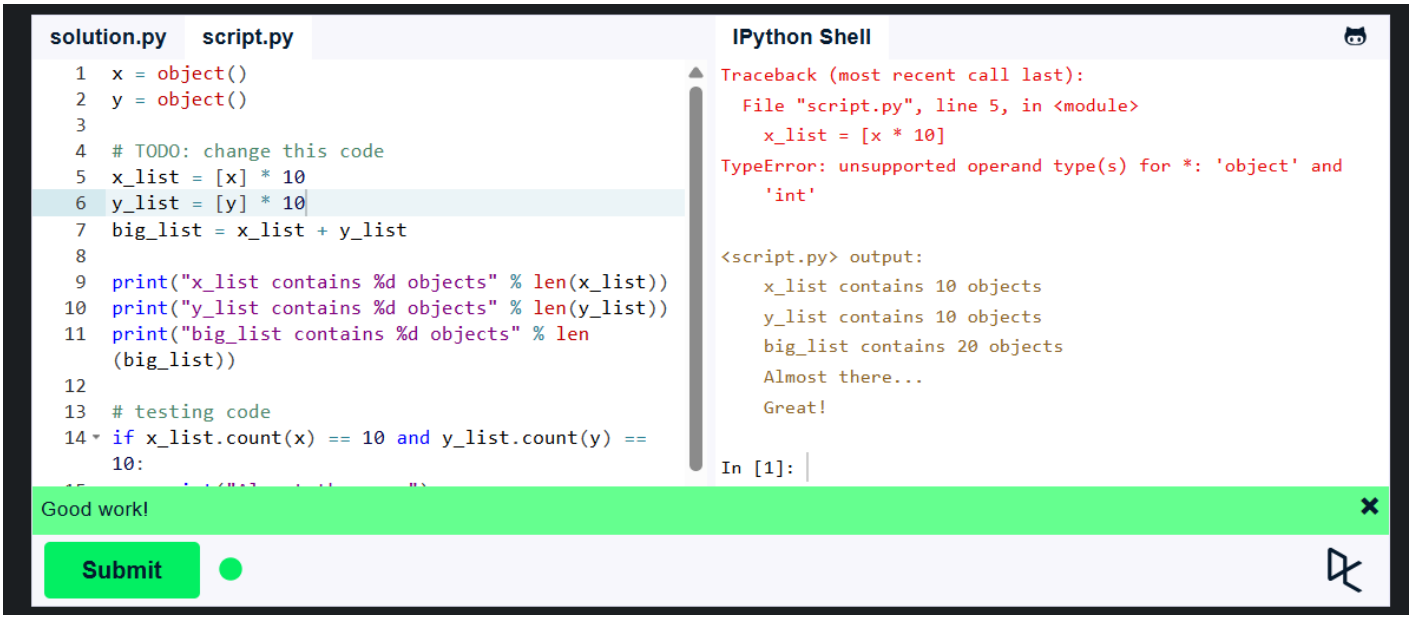


Figura 2.4 – Basic Operators

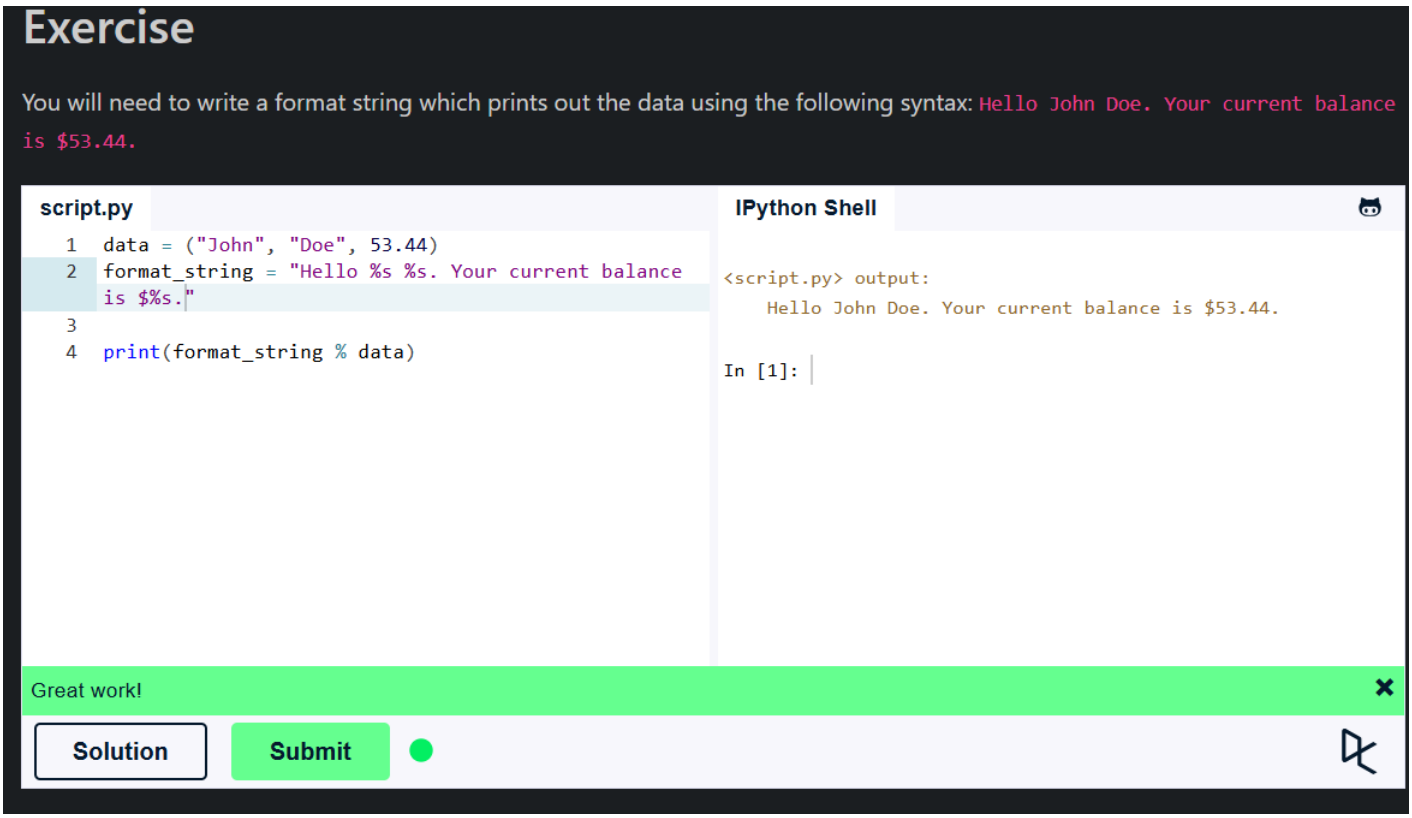


Figura 2.5 – String Formatting

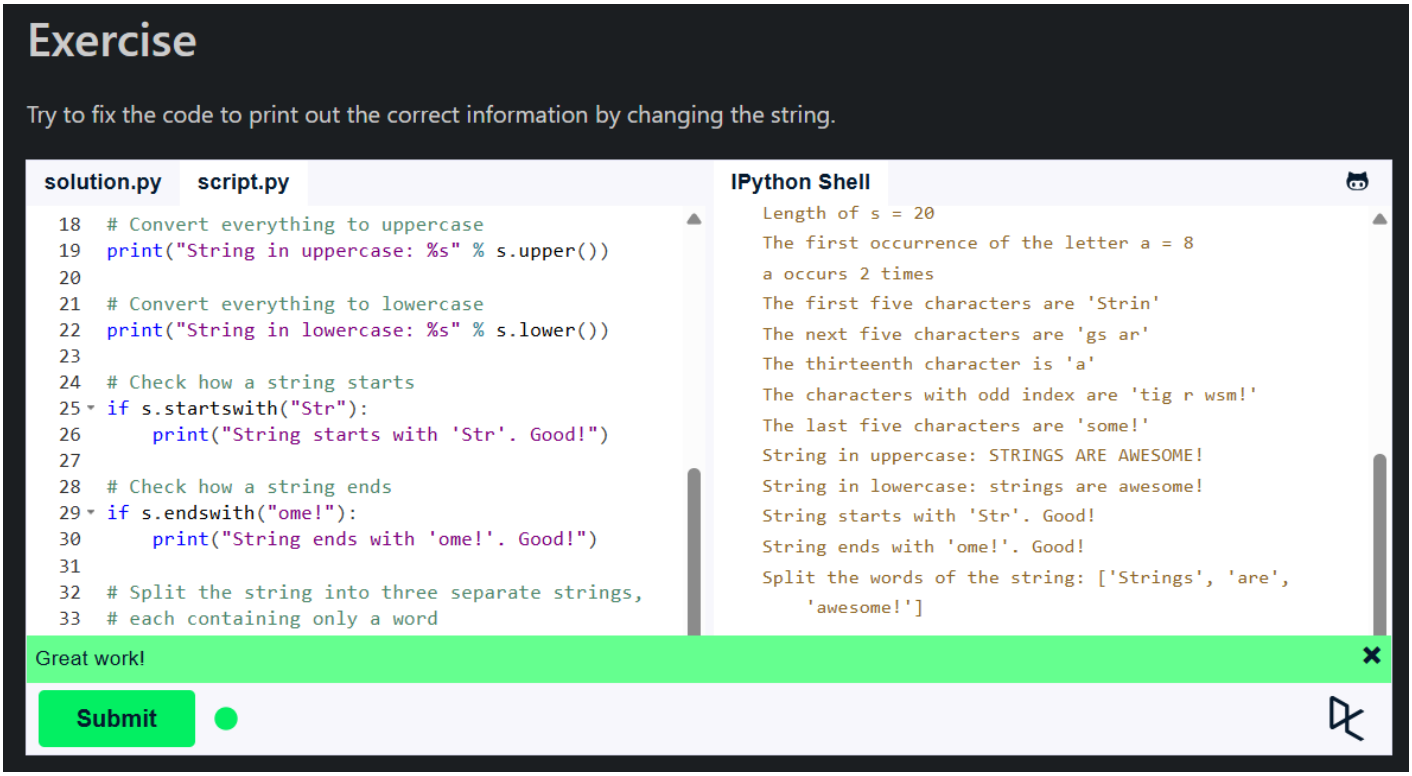


Figura 2.6 – Basic String Operations

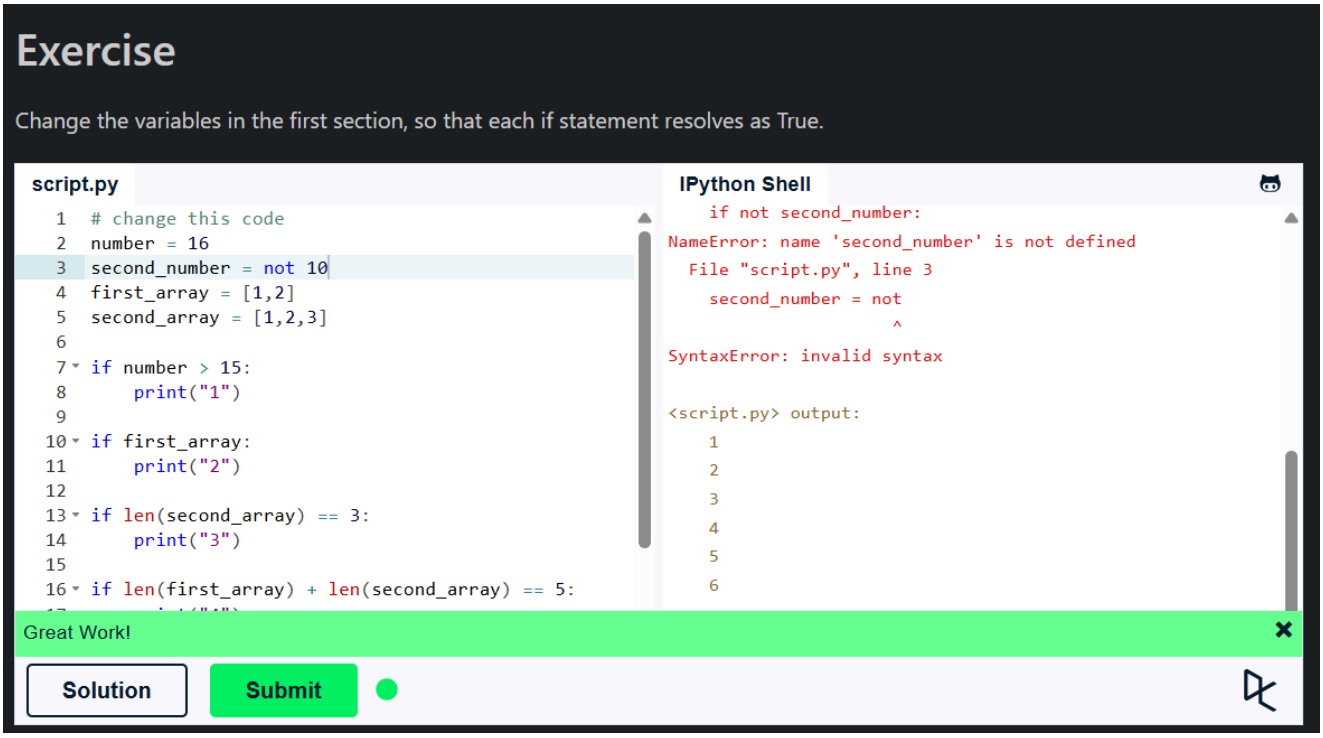


Figura 2.7 – Conditions

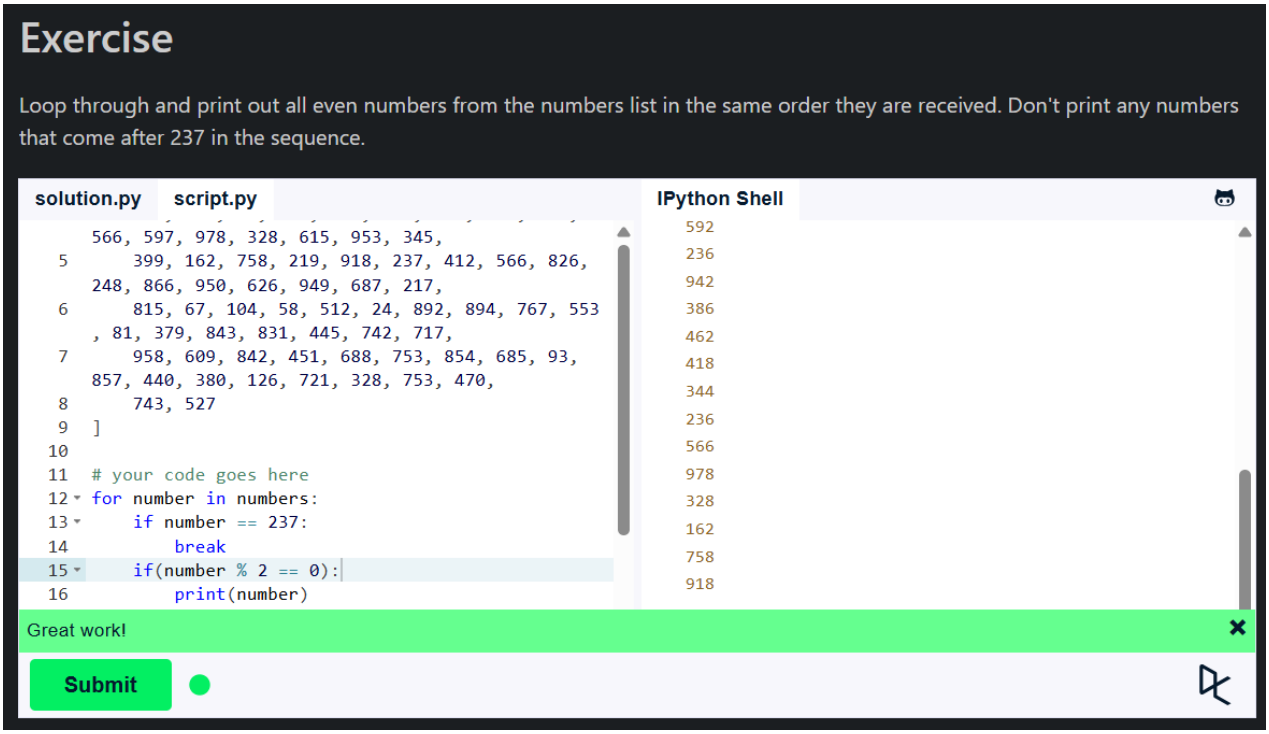


Figura 2.7 – Loops



Figura 2.8 – Functions

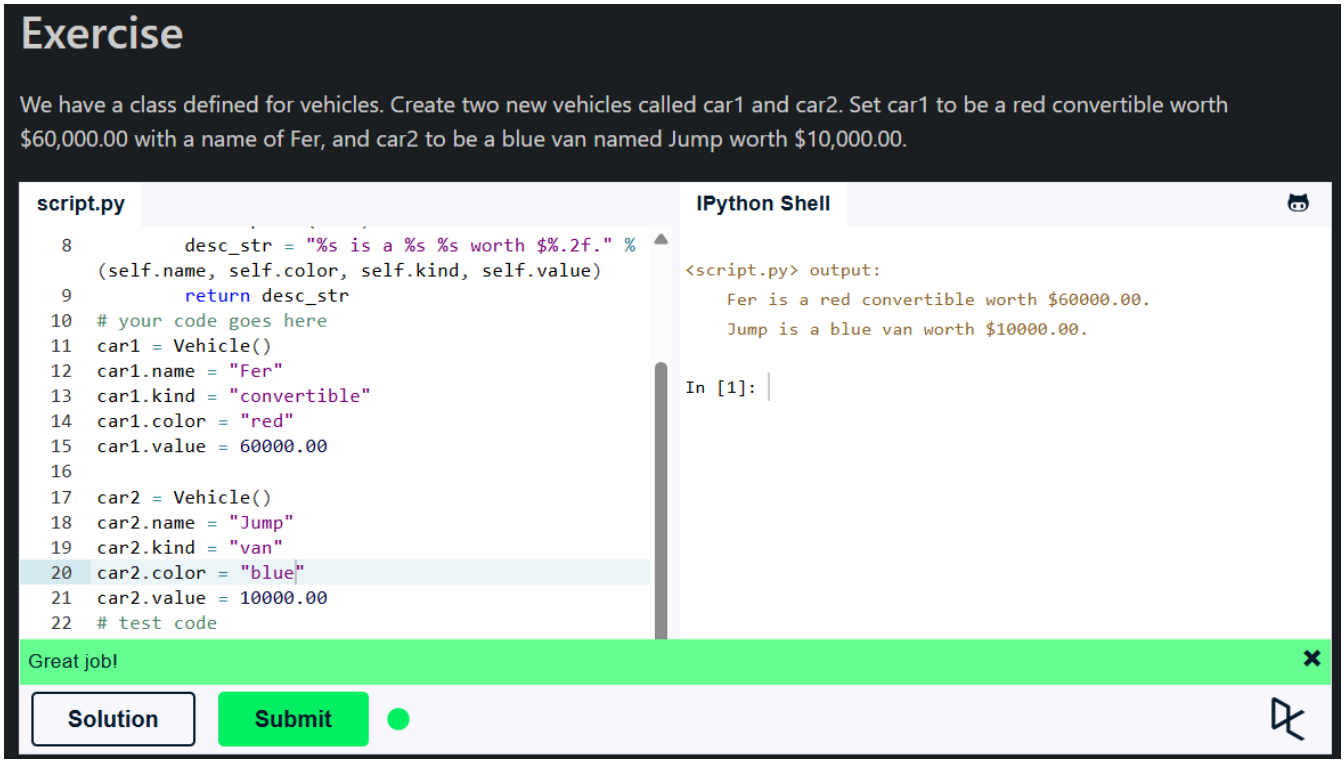


Figura 2.9 – Classes and Objects

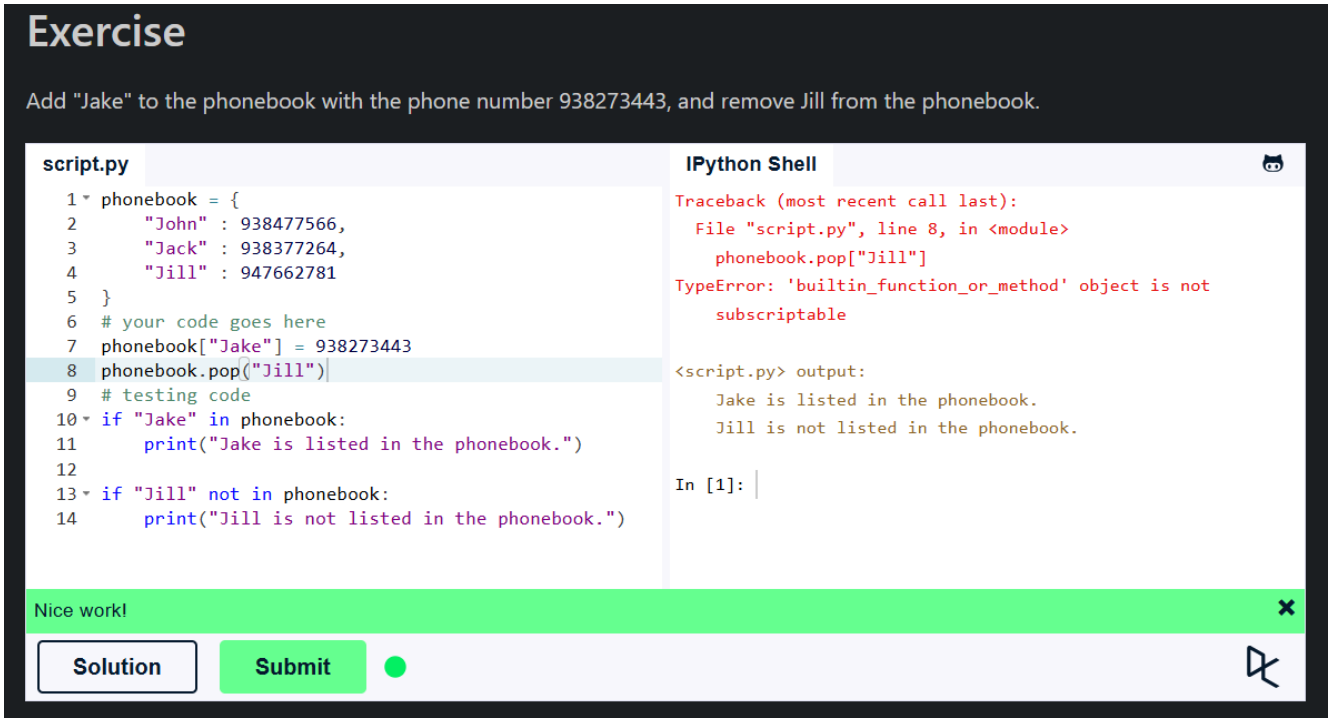


Figura 2.10 – Dictionaries

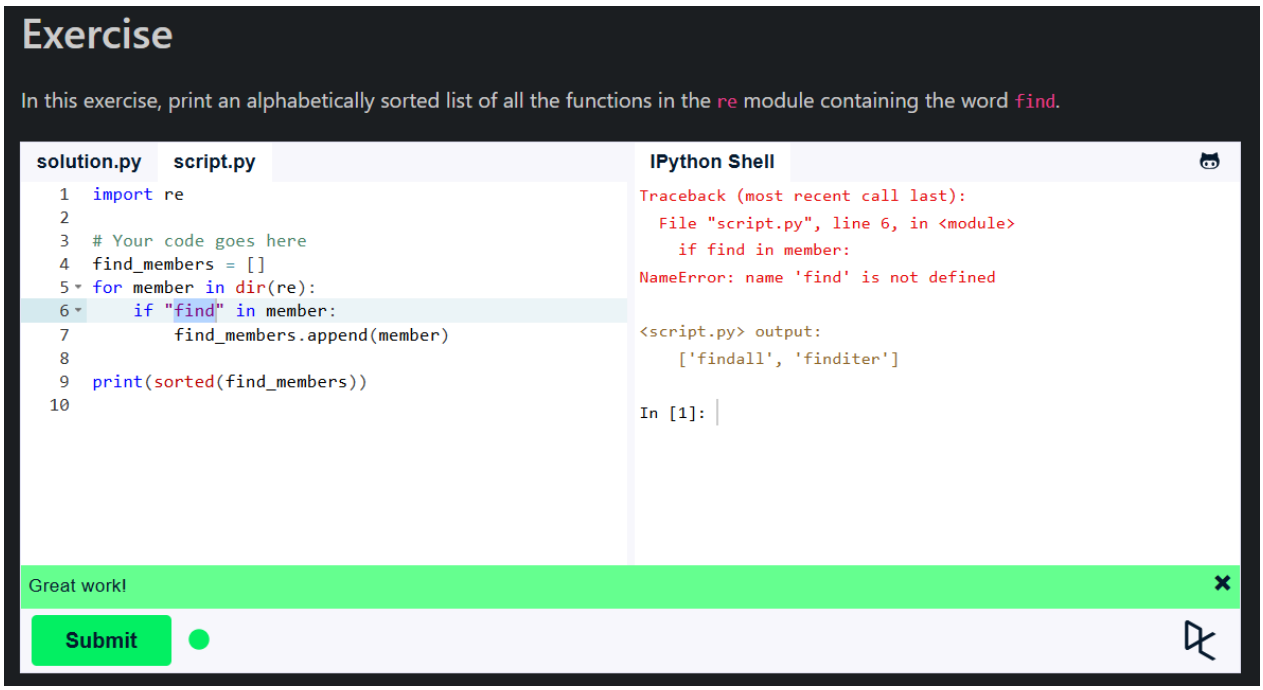


Figura 2.11 – Modules and packages

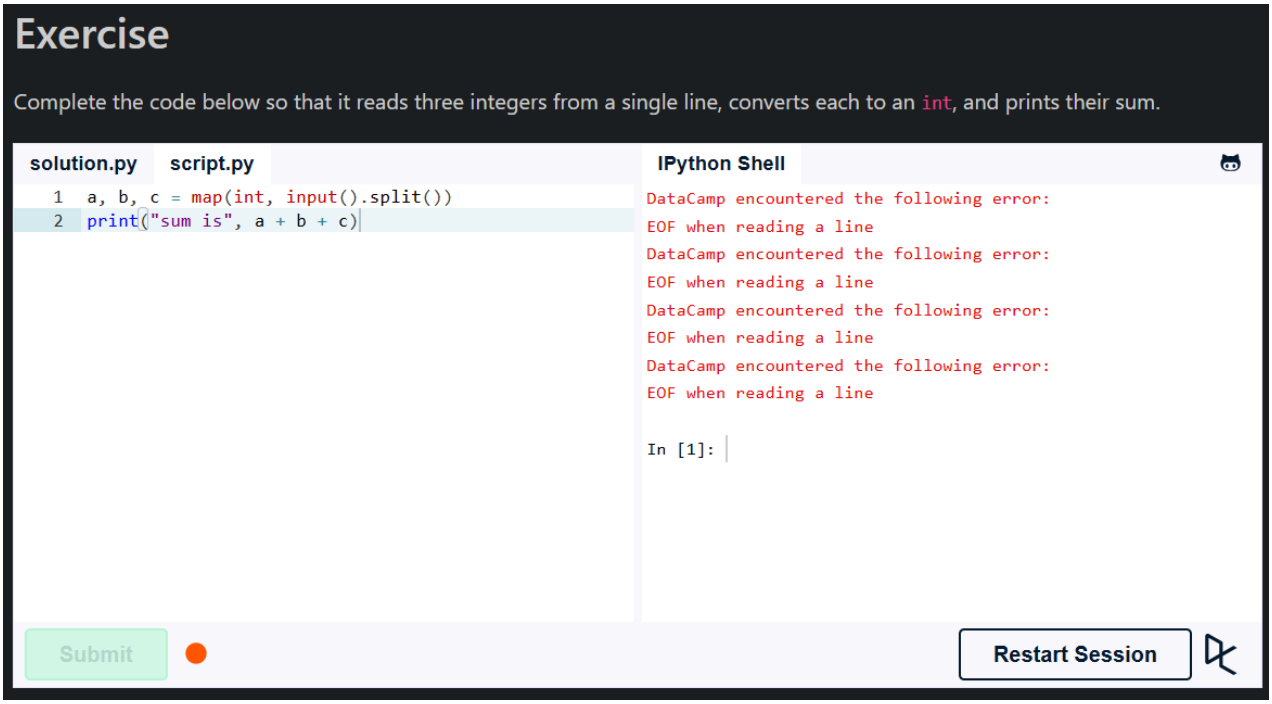


Figura 2.12 – Input and Output

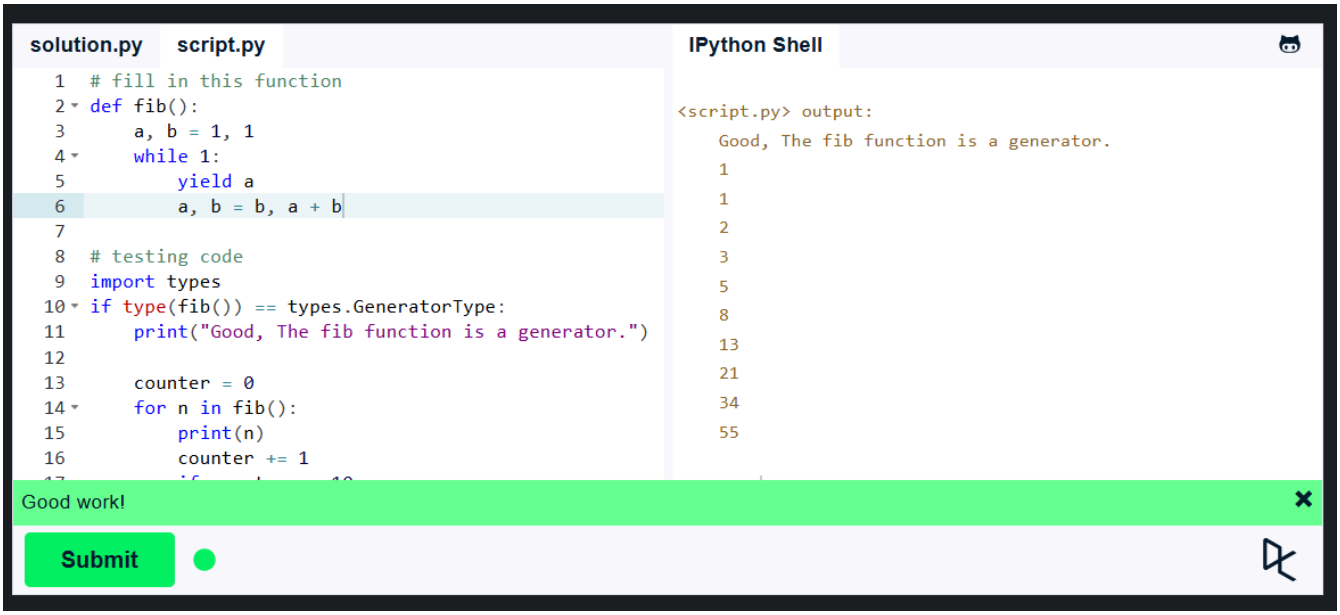


Figura 2.13 – Generators

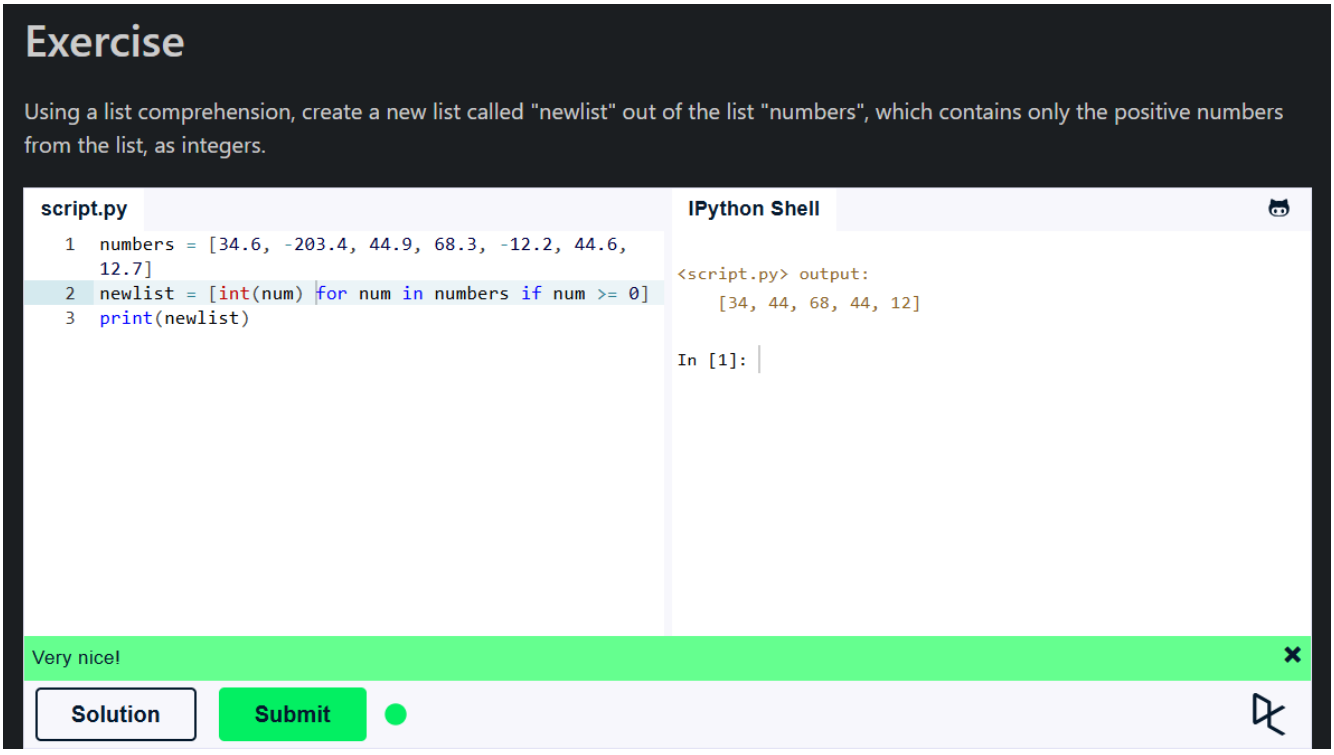


Figura 2.14 – List Comprehension

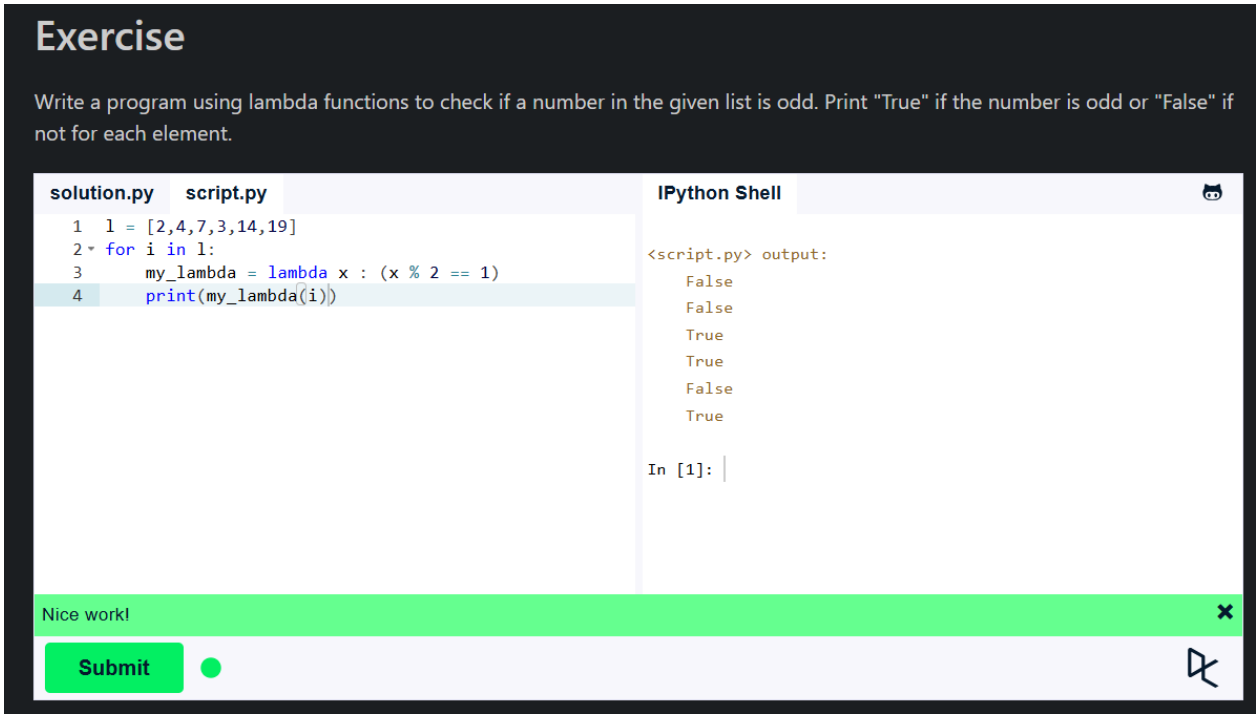


Figura 2.15 – Lambda functions

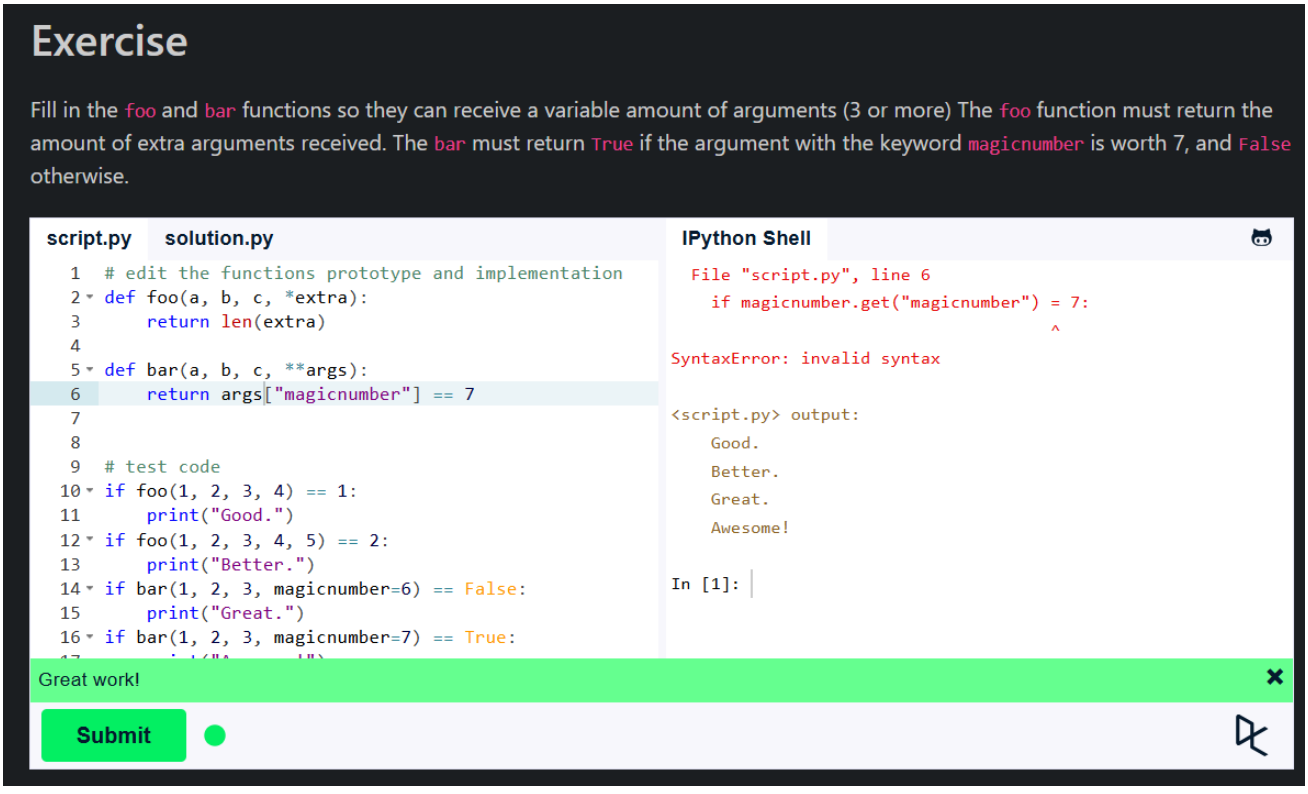


Figura 2.16 – Multiple Function Arguments

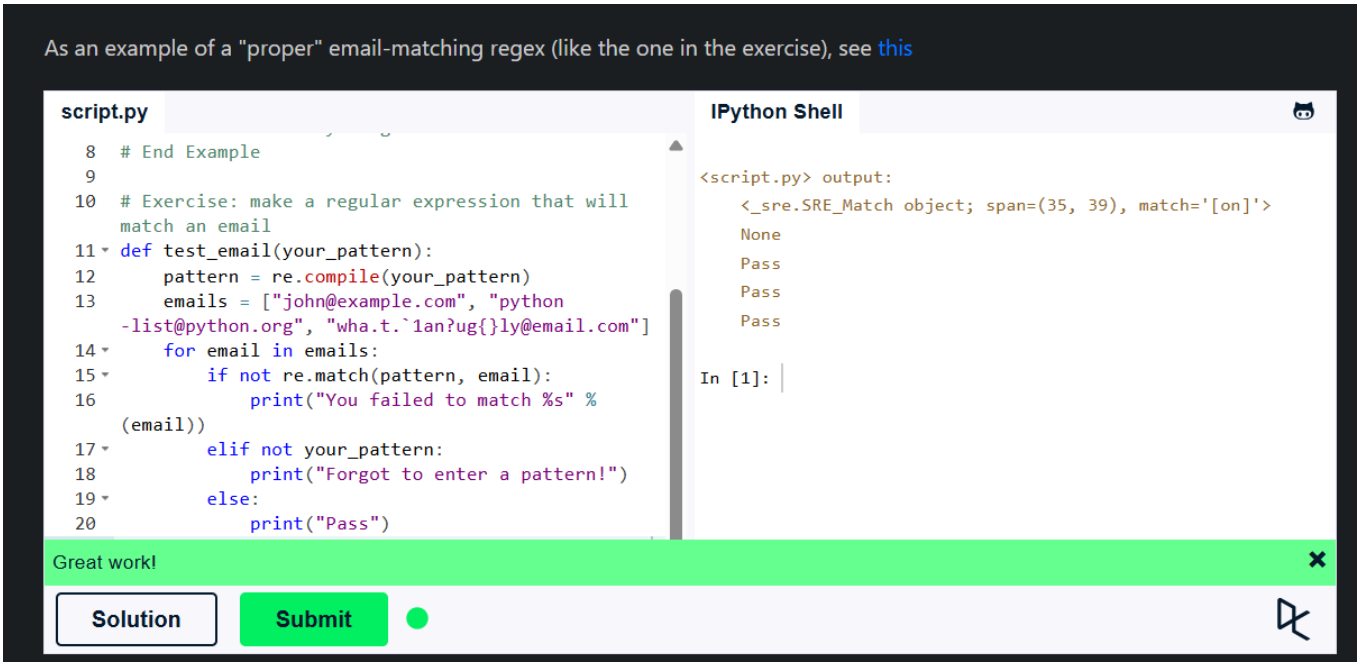


Figura 2.17 – Regular Expressions

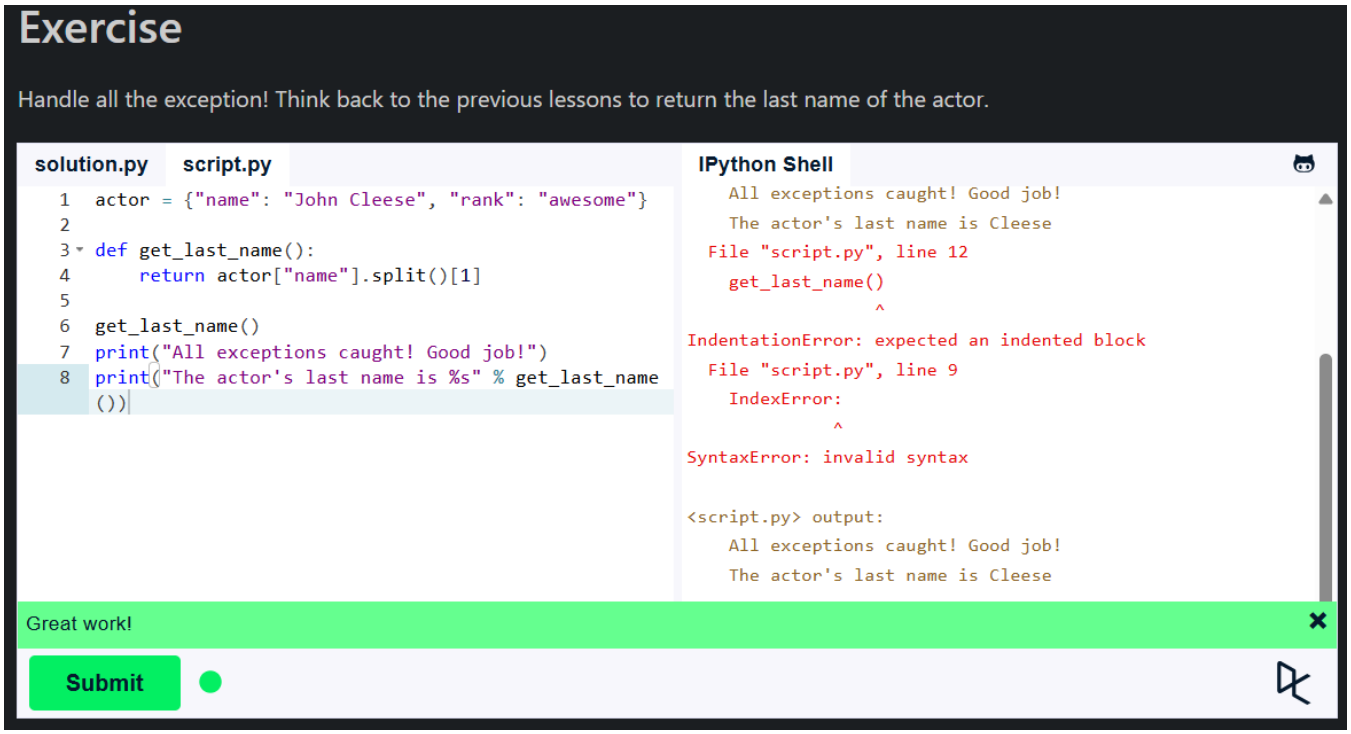


Figura 2.18 – Exception Handleing

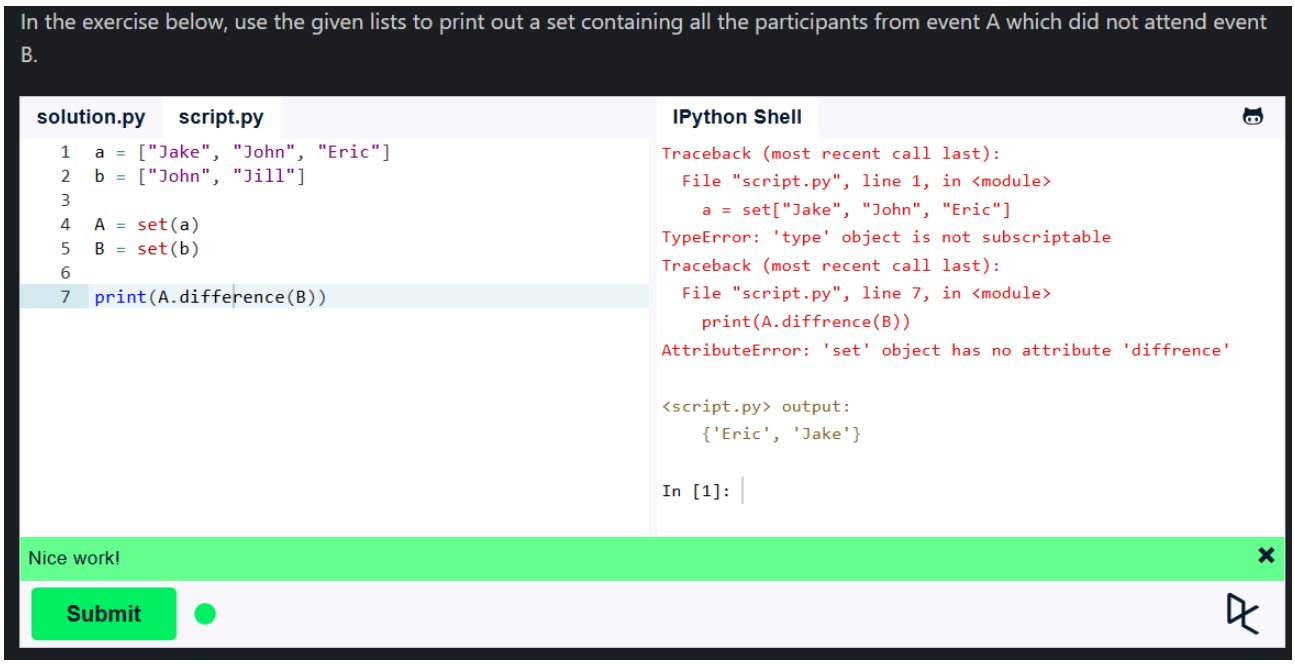


Figura 2.19 – Sets

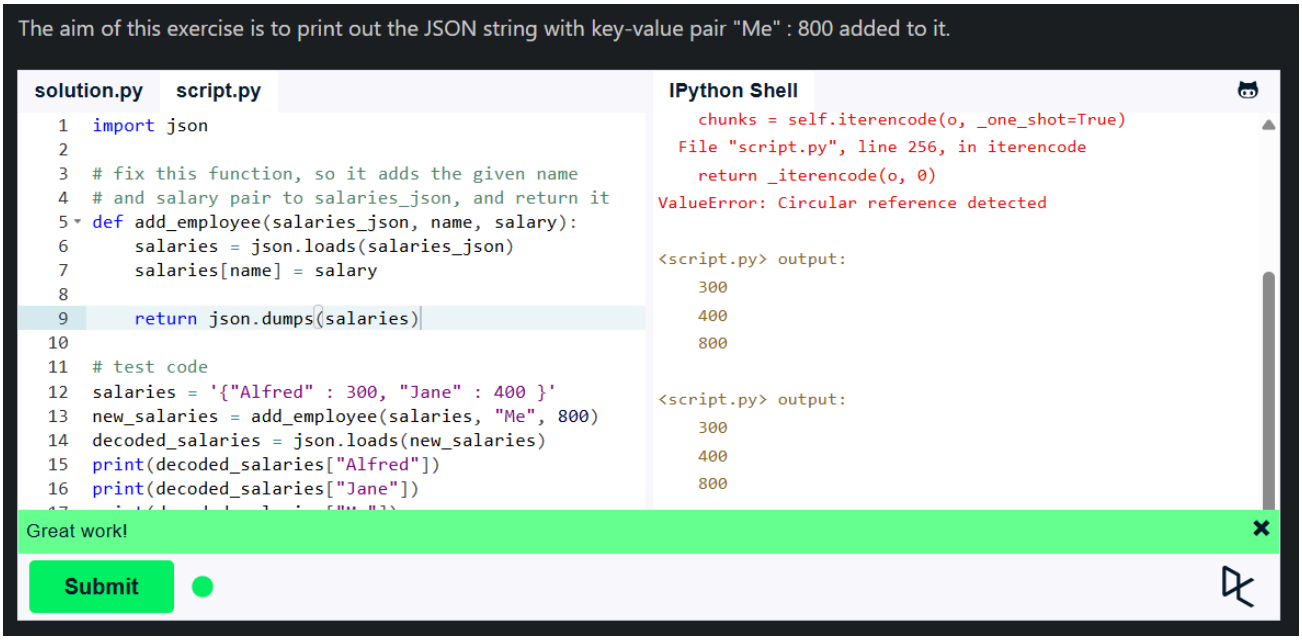


Figura 2.20 – Serialization

CONCLUZII

În această lucrare am învățat Python prin exerciții practice. Am lucrat cu variabile, liste, funcții, clase, excepții și JSON. Platformele interactive m-au ajutat să înțeleg mai ușor sintaxa și să scriu cod corect.

BIBLIOGRAFIE

1. https://www.codedex.io/python
2. https://learnpython.org/